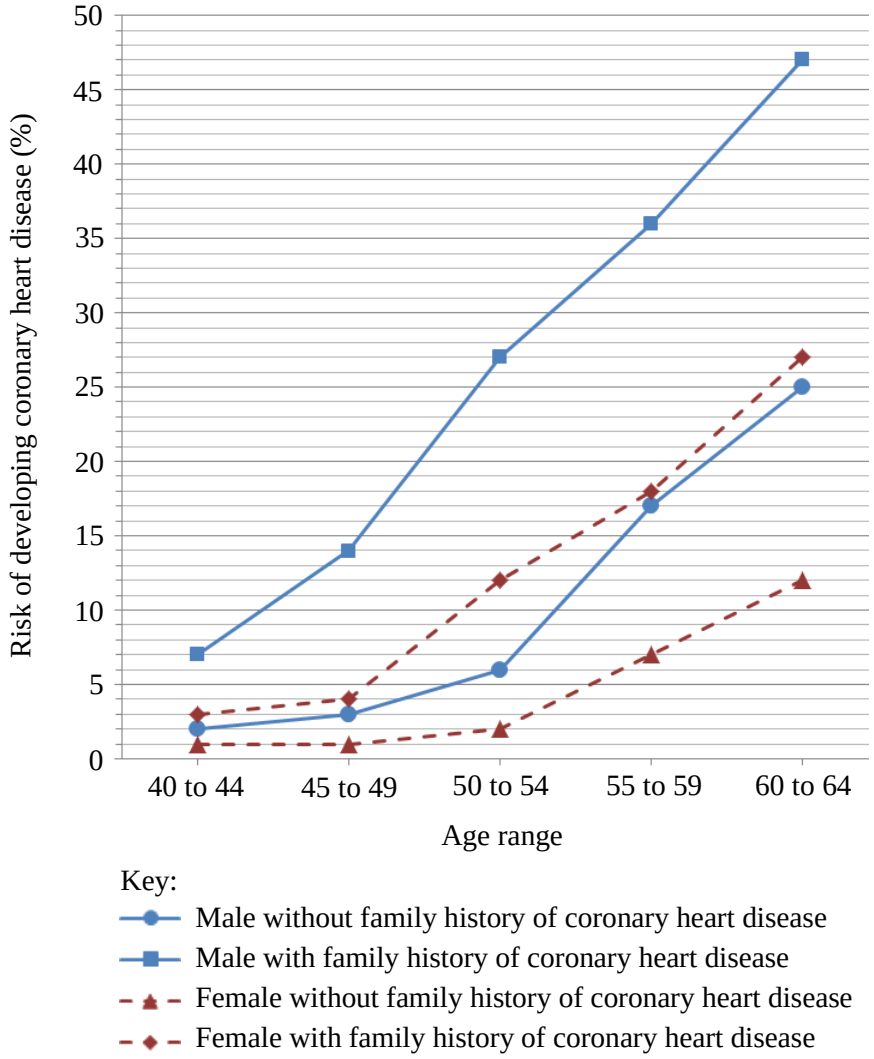
Short questions

|!|BCESA0240000001|!| ★★

Coronary heart disease is common in some families and men are more susceptible to the disease than women. Researchers studied the risk of coronary heart disease in groups of men and women of different ages and with or without family history of the disease. The graph below shows the results.



(a) What is the risk of developing coronary heart disease for a 50-year-old man (without family history) and a 50-year-old woman (without family history), respectively? (2 marks)

© 2020 Aristo Educational Press Ltd.

(b) Calculate the increase in risk for a 60-year-old man with family history of coronary heart disease over a 60-year-old man without such family history. Show your working. (2 marks)

(c) Describe the patterns shown in the graph. (3 marks)

##

(a) 50-year-old man (without family history): 6% (1)
50-year-old woman (without family history): 2% (1)

(b) Increase in risk = 47% – 25% (1)
= 22% (1)

(c) Any ***three*** of the following: (3)

- For all groups, the risk of developing coronary heart disease increases with age.
- For both genders, the groups with family history of coronary heart disease have a higher risk over the groups without such family history.
- Men (without family history of coronary heart disease) have greater risk of developing coronary heart disease than women (without family history of coronary heart disease) at all ages.
- The increase in risk for men with family history of coronary heart disease over men without such family history is greater than the increase in risk for the female groups at all ages.

(Or other reasonable answers)

(7 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2

© 2020 Aristo Educational Press Ltd.

(b)	Mathematical skills	2
(c)	Analysing data	3

|!|BCESB0240000002|!| ★ **STSE**

Colorectal cancer is one of the commonest cancers in Hong Kong. In 2015, there were over 5000 new cases, of which more than 90% of the patients were 50 years of age or older.

(a) Suggest why the incidence of colorectal cancer is higher in persons aged 50 or above. (2 marks)

(b) Give ***two*** other risk factors, other than age, that increase the risk of colorectal cancer. (2 marks)

(c) The Department of Health suggests residents aged 50 or above to receive screening tests for colorectal cancer. Explain the significance of screening programmes in the prevention of colorectal cancer. (1 mark)

##

(a) In older people, their colons may have been repeatedly exposed to carcinogens in food. (1)

Old age allows mutations in the cells to accummulate. (1)

(b) Any ***two*** of the following: (2)

- family history
- diet high in fat and red meat but low in dietary fibre
- lack of physical activity
- alcohol consumption

(Or other reasonable answers)

© 2020 Aristo Educational Press Ltd.

(c) Screening tests allow early detection of any abnormal growth of tissues that may progress into cancer. (1)

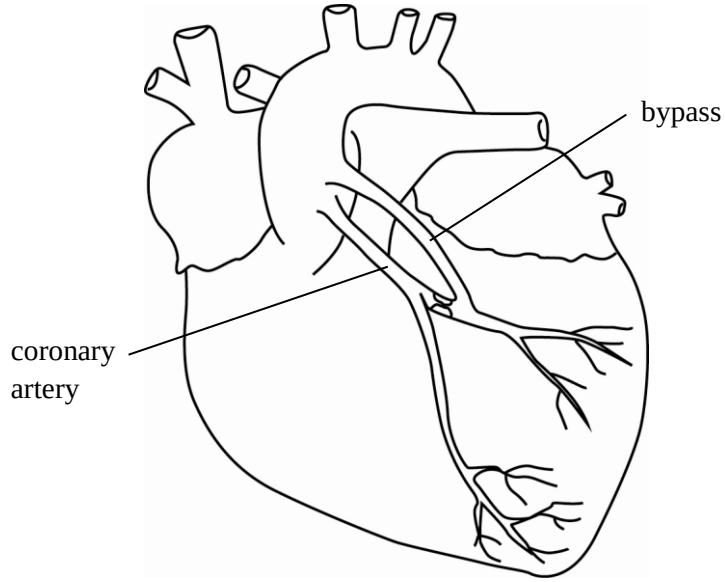
(5 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	2
(b)	Understanding basic concepts	2
(c)	Applying concepts	1

|!|BCESB0240000003|!| ★ **STSE**

The following diagram shows the heart of a person who has had a coronary bypass surgery. The person has suffered from coronary heart disease as a result of plaque formation in a coronary artery.



(a) Describe the process of plaque formation and how it affects the blood flow through the coronary artery. (3 marks)

© 2020 Aristo Educational Press Ltd.

(b) How can a coronary bypass surgery treat coronary heart disease? (3 marks)

(c) Suggest ***one*** way to prevent coronary heart disease. (1 mark)

##

(a) Cholesterol in blood deposits on the inner wall of a coronary artery and forms plaque. (1)
Over time, plaque slowly builds up / becomes larger in size. (1)
The plaque narrows the lumen of the coronary artery and reduces the blood flow to cardiac muscle. (1)

(b) The bypass /grafted blood vessel supplies blood to the affected cardiac muscle. (1)
The cardiac muscle can obtain oxygen and glucose from the blood (1)
for respiration to release energy for their normal functioning. (1)

(c) Any ***one*** of the following: (1)

- Adopt a healthy diet.
- Do exercise regularly.
- Give up smoking.

(Or other reasonable answer)

(7 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Applying concepts	3
(c)	Applying concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCESC0240000004|!|

Liver cancer is one of the common types of cancers in Hong Kong. Drinking alcohol is a risk factor of liver cancer. When the liver metabolizes alcohol, a carcinogen is formed.

(a) Describe how constantly drinking too much alcohol may speed up the process of normal liver cells becoming cancerous. (3 marks)

(b) State ***two*** characteristics of malignant tumours that distinguish them from benign tumours. (2 marks)

##

(a) Everytime the liver breaks down alcohol, the carcinogen that causes mutations is produced. (1)

The liver cells are repeatedly exposed to the carcinogen and mutations in the cells accumulate. (1)

When the genes that regulate cell division undergo mutation, the liver cells divide uncontrollably and excessively, forming a malignant tumour. (1)

(b) Any ***two*** of the following: (2)

- Malignant tumours develop their own blood vessels to obtain more oxygen and nutrients to support their growth.
- Malignant tumour cells may enter the circulatory system, travel to other parts of the body and develop into new malignant tumours.

(5 marks)

##

	Skills assessed	Marks
(a)	Applying concepts	3
(b)	Applying concepts	2

© 2020 Aristo Educational Press Ltd.

|!|BCESC0240000005|!|

Mr Wong is a smoker and loves eating meat and other high-fat foods. One day, he collapsed on the street. He was sent to the hospital and was diagnosed with stroke.

(a) Explain briefly what stroke is. (2 marks)

(b) Explain how smoking and high-fat diet may lead to stroke.

(i) smoking (2 marks)

(ii) high-fat diet (3 marks)

##

(a) Stroke occurs when an artery in the brain is blocked or bursts. (1)
The blood supply to the brain cells nearby is cut off and the brain cells die due to lack of oxygen. (1)

(b) (i) Nicotine in cigarette smoke increases blood viscosity and the chance of forming blood clots (1)
which may block the narrowed blood vessels. (1)

(ii) A high fat intake from the diet raises the blood cholesterol level. (1)
Excess cholesterol in blood is more likely to deposit on the inner wall of an artery in the brain. (1)
The lumen of the artery becomes narrower and may finally become completely blocked. (1)

(7 marks)

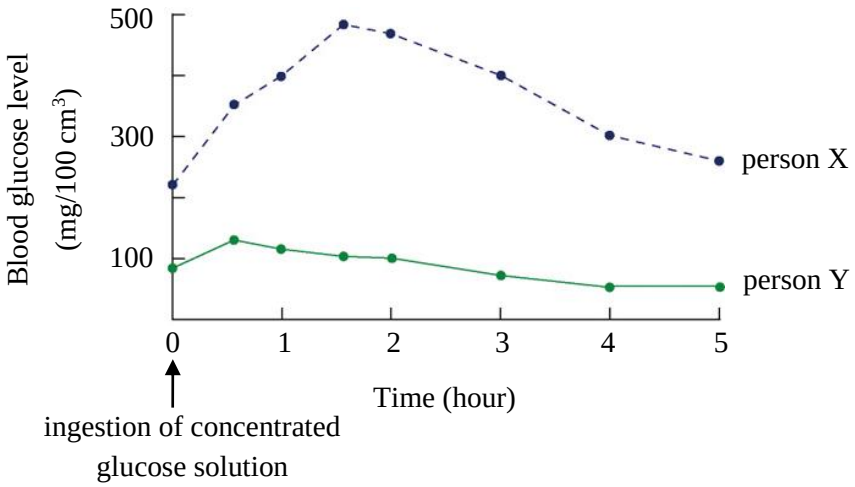
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)(i)	Applying concepts	2
(b)(ii)	Applying concepts	3

© 2020 Aristo Educational Press Ltd.

|!|BCESB0240000006|!| ★

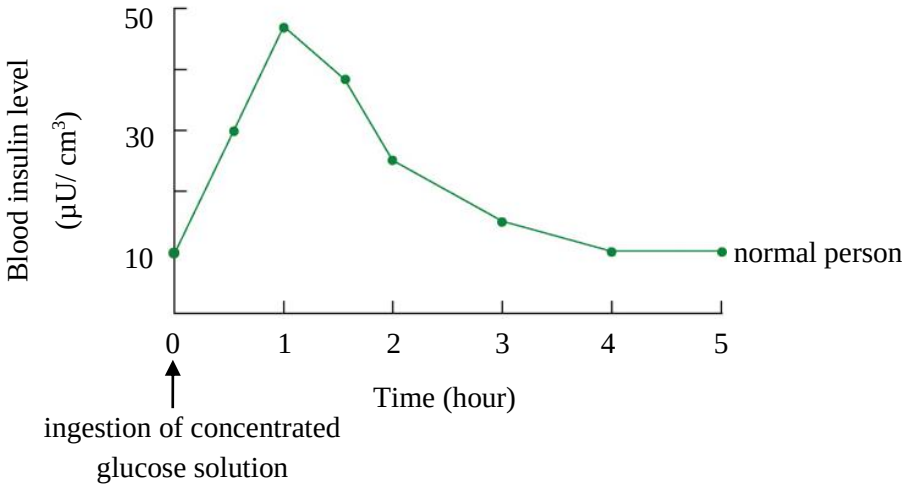
In a medical test, two persons (X and Y) were fasted overnight and then each drank 200 mL of concentrated glucose solution. Their blood glucose levels were monitored for five hours. The graph below shows the changes recorded.



(a) Which of the two persons is diabetic? Explain your answer. (4 marks)

(b) It is known that the diabetic person has Type 1 diabetes.
The graph below shows the changes in the blood insulin level of the normal person. Sketch a curve to show how the blood insulin level of the diabetic person would change. (2 marks)

© 2020 Aristo Educational Press Ltd.



(c) What treatment should a patient with Type 1 diabetes receive? (1 mark)

##

(a) Person X has diabetes. (1)

Before the ingestion of glucose solution, person X's blood glucose level was much higher than that of person Y. (1)

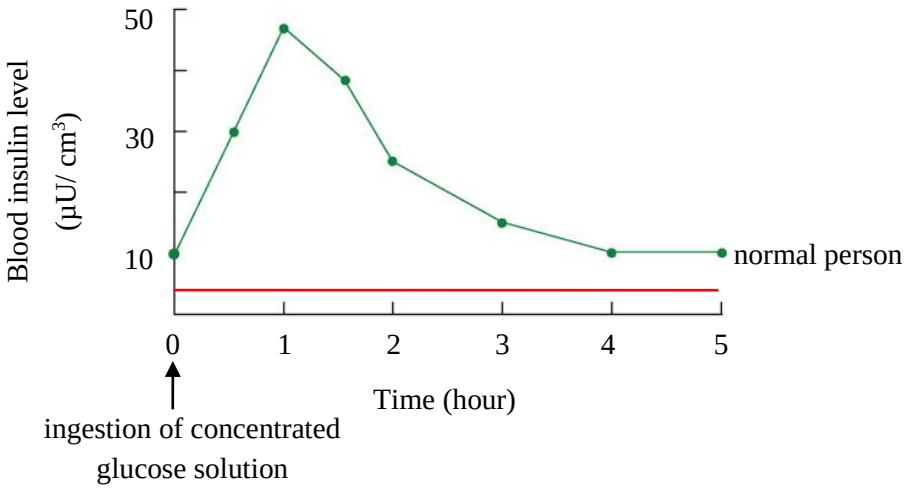
After the ingestion of glucose solution, person X's maximum blood glucose level was five times higher than that of person Y. (1)

It took much more time for person X's blood glucose level to return to its initial level compared to person Y. (1)

(b) The blood insulin level of the diabetic person would remain low even after the ingestion of glucose solution. (1)

The blood insulin level of the diabetic person would be lower than that of the normal person. (1)

e.g.



(c) Receive regular insulin injections (1)

(7 marks)

© 2020 Aristo Educational Press Ltd.

##

	Skills assessed	Marks
(a)	Analysing data	4
(b)	Applying concepts	2
(c)	Understanding basic concepts	1

|!|BCESC0240000007|!|

For each of the cancers listed in ***Column 1***, select from ***Column 2*** the most relevant risk factor. Put the appropriate letter in the space provided. (4 marks)

Column 1			***Column 2***
Colorectal cancer	(a)	_____	A. smoking
Lung cancer	(b)	_____	B. diet high in fats and low in dietary fibre
Skin cancer	(c)	_____	C. infection of human papilloma virus
Cervical cancer	(d)	_____	D. infection of *Helicobacter pylori*
			E. prolonged exposure to ultraviolet light

##

(a) B (1)
(b) A (1)
(c) E (1)
(d) C (1)

_____
(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.

|!|BCESB0240000008|!| ★

For each of the types of non-infecticous diseases listed in ***Column 1***, select from ***Column 2*** the substance(s) in cigarette smoke that increase(s) the risk of disease development. Put the appropriate letter in the space provided. (3 marks)

Column 1		***Column 2***
Lung cancer	(a) _____	A. nicotine
Cardiovascular diseases	(b) _____	B. tar
		C. carbon monoxide

##

(a) B (1)

(b) A, C (1, 1)

(3 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1

© 2020 Aristo Educational Press Ltd.